Sheaf-cohomology QC for volume-EM registration: pre-registered negative results and an exact view-gauge invariance

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Abstract

We operationalized cohomological inconsistency (failure of adjacent-frame displacement measurements to form a 1-coboundary on the coverage nerve) as a quality-control detector for volume-electron-microscopy registration, and evaluated it, pre-registered, on three real channels: two distant ROIs of an EMPIAR-10478 HeLa SBF-SEM block and one vendor-registered EMPIAR-10070 FIB-SEM muscle stack. Three findings survive. (1) An **exact view-gauge invariance** (an elementary cancellation identity, stated as a remark below): whole-view displacement is an exact gauge invariance of relative-content measurement; the contribution is empirical — a shift-channel anomaly scores AUROC exactly 0.500 at every amplitude on real phase-correlation measurements. (2) **Detectable bias is empirically detector-saturated within the tested closure-residual family**: an injected edge defect crosses AUROC 0.9 only at ∼4–5× the triangle-closure noise scale on raw SBF (GLR-optimal and max-abs detectors differ by < 0.03; a re-implemented pairwise photometric QA baseline scores *lower* than the sheaf channel at matched relative scale), while pre-registered FIB-SEM residuals sit ∼15× lower in absolute pixels yet still admit no $\leq 3\times$ detection. A threshold sweep of the photogrammetry prune rule (1.5–4×RMS) finds it budget-driven, not rule-driven: a workable τ differs by channel (2.5× SBF vs 3× FIB-SEM at 1% false flags) and no universal multiple transfers — scoped to single-edge offset defects at K=6; global bundle-adjustment pruning is not tested here. and closure-σ scales span two orders of magnitude across techniques: QC thresholds must be absolute per-task tolerances, never scale-free noise multiples. (3) An **instrument-disclosure protocol**: a self-referential-anchor defect that flatlined the first detector at chance was caught by an algebraic gauge proof before any gate result was seen. All results, including nulls, are frozen in git before each run; code and artifacts are public at https://github.com/ScivizLabVienna/e3-sheaf-data.

1 Introduction

Large volume-EM reconstructions are assembled by registering thousands of adjacent section images; registration errors propagate silently into segmentation and connectomics. Existing automated quality assessment for this step is learned and pairwise — optical-flow SIQA metrics [4, 3], learned features [5], vision-language judges [6] — and carries no structural statement about *which* defects a given statistic can or cannot see.

Cellular sheaves on coverage nerves offer exactly such a statement: displacement measurements on overlaps are a 1-cochain, a consistent registration is a coboundary, and the closure residual of every triangle is a per-edge-usable inconsistency signal with no learned parameters. Prior registration work uses the same formalism to *eliminate* loop error globally (Poisson/Hodge solving) [2], never to *detect* per-edge defects against a noise-calibrated threshold.

We adopted the minimal instance of this detector, pre-registered acceptance gates, and ran it on real archives. The headline outcome is negative and useful: within the tested statistics, what the detector can see is set by the channel, not by detector cleverness (we do not rule out statistics outside the closure-residual family), and the two structural claims below are exact.

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2 Formal setting

Frames $0, \ldots, N-1$ form a path nerve; each adjacent pair $(j, j+1)$ carries a measured translation $m_{jj+1} \in \mathbb{R}^2$ (phase correlation on overlapped ROIs). A *closure window* of K consecutive edges forms triangles whose residual is $\rho = m_{12} + m_{23} - m_{13}$ (global-reference legs measured independently at wide baseline). Two defect classes: **M_edge** — a wrong offset on one edge (e.g. a bad correspondence patch); **N_shift** — a whole-view translation desync.

> **Remark (view-gauge invariance).** *Adding a per-view translation t_j to all measurements touching view j leaves every closure residual invariant; hence no coboundary-channel statistic can discriminate an N_shift at any amplitude.* This is an elementary cancellation identity, not a new structural result; its value here is that it predicts and explains an exactly-0.500 empirical null, which no learned metric can.

Proof: residuals are alternating sums in which each t_j appears with coefficients summing to zero. The empirical instance is exact, not approximate (§4).

> **Proposition (redundancy dilution).** *In a window of K edges solved by weighted least squares, one corrupted edge's offset contributes to per-row residuals at $O(1/K)$; detection difficulty grows with the very redundancy that improves fit quality.*

3 Methods (frozen before each run)

Instrument (E6b): independent global-reference legs by wide-baseline (+30-frame) phase correlation plus adjacent lag-10 rows; triangle-closure residuals define σ_c per channel, emitted *before* any injection trial. Trials: $K = 6$, 40 trials per class, defect levels $\{0.5, 1, 2, 3, 5\} \times \sigma_c$; detectors max-abs residual and GLR-optimal whitened χ^2 energy [1]. Gate (identical for E6/E7/E8/E9): AUROC ≥ 0.9 at level $\leq 3 \times \sigma_c$, else detector-saturation verdict for that channel. Channels: EMPIAR-10478 SBF ROI1 (576,297,858 B tile, 151×2000×2000), ROI2 (740,048,035 B), EMPIAR-10070 FIB-SEM MITOMAP stack [7] (777,443,899 B, 291×1595×1675, vendor TTrAC-registered). A synthetic campaign (P1–E5, oracle conditions) closed the space before real data.

4 Results

Table 1: Closure σ_c and GLR AUROC at $\{0.5, 1, 2, 3, 5\} \times \sigma_c$ (bootstrap CIs in artifacts). N_shift: 0.500 exactly at all levels (ROI1).

Channel	σ_c (px)	0.5×	1×	2×	3×	5×
SBF ROI1	1.78 / 1.66	0.600	0.720	0.844	0.890	0.943
SBF ROI2	1.46 / 1.47	0.593	0.696	0.823	0.881	0.927
FIB-SEM	0.11 / 0.14	0.571	0.586	0.628	0.666	0.737

Table 4 and Figure 4 report the level curves. No channel meets the gate at $\leq 3\times$; both SBF ROIs cross 0.9 only between 3× and 5× (E8 replicates E6/E7 on a distant ROI). GLR energy beats max-abs by $\leq +0.025$ AUROC everywhere; a re-implemented pairwise photometric QA baseline (EMSIQA/OFSIQA recipe [4, 3]; 360 clean ROI1 pairs) scores *worse*, AUROC 0.53 at 3×. Detector design adds little *within the tested family*; statistics outside it remain open. FIB-SEM inverts the mechanism — vendor pre-registration drives σ_c two orders of magnitude down, and redundancy dilution (Proposition) plus absolute-scale mismatch leave AUROC <0.9 even at 5×. *Closure-σ self-calibration does not normalize detection across channels.*

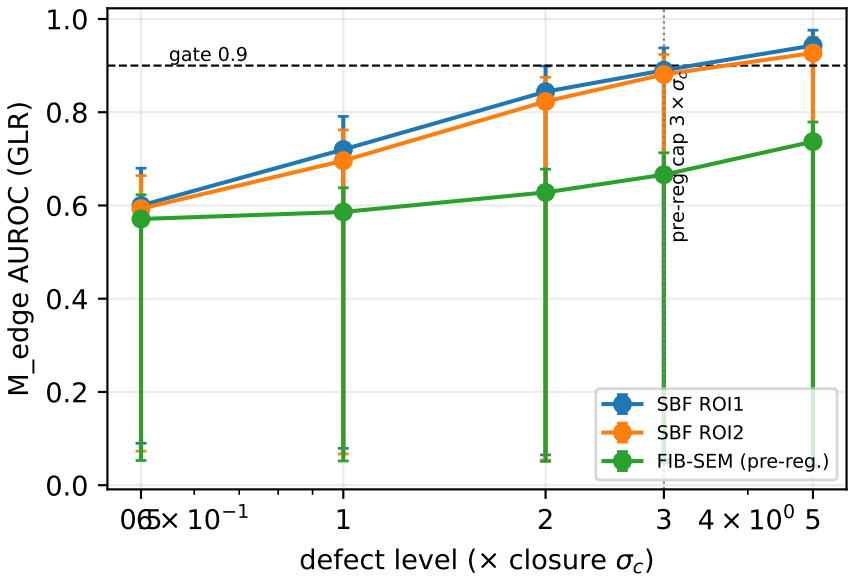


Figure 1: MDB level curves for the injected M_edge defect on all three channels (GLR detector; markers from the committed result artifacts, error bars are the pre-registration-frozen bootstrap CIs). The frozen gate (0.9 at $\leq 3\times$) is missed everywhere; SBF crosses between $3\times$ and $5\times$, FIB-SEM has not reached 0.9 by $5\times$.

5 Discussion

Operational rule. The field practice of pruning registration residuals above 2–3×RMS [10] implicitly assumes noise-dominated channels. Our threshold sweep (GLR energy, hard flag at $\tau\times$null-RMS, $\tau \in \{1.5, \ldots, 4\}$, 86/120/226 calibration windows per channel) shows the practice is *budget-driven, not rule-driven*: a 2–3× prune catches 53–74% of injected $\leq 3\times$ SBF defects but only at a 0.8–4.7% false-flag rate (SBF defect residuals sit in the far chi^2 tail; null RMS badly underestimates it), while the same rule flags only 19–30% of FIB-SEM defects at a comparable budget and needs $\tau\approx 1.5\times$ just to reach 13% false flags versus $\tau \geq 2.5\times$ on SBF: no universal τ transfers, and the operative dial is the tolerated false-flag rate, not a noise multiple. Thresholds must be absolute per-task tolerances. The same caveat applies to calibration of learned QA metrics [4, 6].

Disclosure protocol. The first E6 instrument contained a self-referential-anchor defect that made the tested statistic an exact gauge by construction (AUROC flat 0.500). It was diagnosed algebraically and disclosed *before* any gate verdict; the rebuilt instrument (independent global legs) reproduced theory. We report this as a methodological contribution for agent-run science.

Limitations. One FIB-SEM stack (no ROI variety); no independent ground truth on EMPIAR-10070 (shifts baked by NEWSTACK; extended header byte-verified history-only), so the oracle-ceiling comparison was run only on SBF legs; serial-section TEM and array tomography untested. σ_c on the FIB-SEM channel may reflect TTrAC smoothing rather than intrinsic channel noise (single stack, n=1); a synthetic-deformation ground-truth stack is queued to separate the two.

6 Related work

Hodge/Poisson elimination of loop error in global registration [2]; EM stitching QA: EMSIQA [4], OF-SIQA/DEMIS [3], learned features [5], VLM judges [6]; sheaf-inconsistency mathematics: consistency radius [8]; streaming H^1 [9]. None evaluates coboundary residuals as calibrated per-edge QC, and none reports pre-registered negative results for volume EM.

7 Reproducibility

Git-campaign record (every gate dated before its run), frozen evaluators, result artifacts and bootstrap CIs: https://github.com/ScivizLabVienna/e3-sheaf-data; source data by byte-size-verified EMPIAR accessions. Rebuild: `e6b_measure.py` → `e7_glr.py` (numpy/scipy/tifffile).

References

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